AI Digital Twin Project Documentation

Problem Statement #10: Build Your AI Digital Twin

Executive Summary

This project creates a realistic AI version of myself that can speak in my own voice through a comprehensive workflow combining image generation, voice cloning, and lip-synced animation. The final deliverable is a 1-minute talking-head video featuring AI-generated images of myself, cloned voice from my recordings, and natural lip-synced animation.

Project Objective

To create a realistic AI digital twin that can speak in my own voice by integrating:

- **Image Generation**: FLUX LoRA training on personal photos
- **Voice Cloning**: Using Chatterbox for voice replication
- **Animation**: InfiniteTalk for lip-sync and facial animation
- **Workflow Integration**: ComfyUI for seamless pipeline execution

Technical Architecture

Core Technologies Used

1. **ComfyUI**: Visual workflow platform for AI pipelines
2. **WanVideo 2.1**: Image-to-video model (14B parameter, 480p)
3. **InfiniteTalk**: Multimodal talking head generation
4. **Wav2Vec2**: Chinese audio processing model
5. **MelBandRoFormer**: Vocal separation for clean audio
6. **CLIP Vision**: Image embedding for consistency
7. **UMT5-XXL**: Text encoding for prompt conditioning

Final Deliverables

- One 1-minute talking-head video
- Exactly four AI-generated images of myself
- Voice cloned from my own recordings
- Lip-synced animated images

All visuals and audio are generated from my own data. No stock avatars or celebrity likenesses were used.

Model Files & Dependencies

Required Model Downloads

Main Models:

- **GGUF Format** (Recommended for lower VRAM):
  - Main: wan2.1-i2v-14b-720p-Q8_0.gguf
  - Location: models/diffusion_models/
  - Source: https://huggingface.co/city96/Wan2.1-I2V-14B-480P-gguf

InfiniteTalk Models:

- MultiTalk: Wan2_1-InfiniTetalk-Single_fp16.safetensors
- Location: models/diffusion_models/InfiniteTalk/
- Source: https://huggingface.co/Kijai/WanVideo_comfy/tree/main/InfiniteTalk

VAE Model:

- File: Wan2_1_VAE_bf16.safetensors
- Location: models/vae/

LoRA:

- File: lightx2v_I2V_14B_480p_cfg_step_distill_rank128_bf16.safetensors
- Location: models/loras/

Audio Models:

1. Wav2Vec2: wav2vec2-chinese-base_fp16.safetensors
   - Location: models/wav2vec2/
   - Source: https://huggingface.co/Kijai/wav2vec2_safetensors
2. MelBandRoFormer: MelBandRoformer_fp16.safetensors
   - Location: models/diffusion_models/
   - Source: https://huggingface.co/Kijai/MelBandRoFormer_comfy

CLIP Vision:

- File: clip_vision_h.safetensors
- Location: models/clip_vision/

Text Encoder:

- File: umt5-xxl-enc-bf16.safetensors
- Location: models/text_encoders/

Workflow Execution Steps

Step 1: Preparation

1. Collect 10-20 high-quality photos of yourself
2. Train FLUX LoRA using your photos (separate process)
3. Record 2-3 minutes of voice samples
4. Write script for the video content

Step 2: Setup ComfyUI

1. Install ComfyUI and all custom nodes
2. Download all required models
3. Place models in correct directories
4. Load the workflow JSON file

Step 3: Configure Input

1. Load your reference image
2. Load your audio recording
3. Adjust text prompt if needed
4. Set max_frames based on desired length

Step 4: Verify Settings

- Width: 640 pixels
- Height: 640 pixels
- Max frames: 500 or audio length (1 Sec = 25 Frames)
- Frame rate: 25 FPS

Step 5: Generate

1. Queue the workflow
2. Monitor VRAM usage
3. Wait for completion (time depends on hardware)
4. Video saves automatically to output folder

Step 6: Review & Iterate

- Check lip-sync accuracy
- Verify facial consistency
- Adjust audio_scale if needed

Workflow Diagram

```
Input Image → Resize (640×640) → CLIP Encoding → Image Embeddings
                                  ↓
                                I2V Conditioning
                                  ↓
Input Audio → Vocal Separation → Wav2Vec2 → MultiTalk Embeddings
                                  ↓
Text Prompt → UMT5 Encoding → Text Embeddings → Sampler → Latents
                                  ↓
                                Decode (VAE)
                                  ↓
                                Frames
                                  ↓
                            Video Combiner + Audio
                                  ↓
                                Final MP4
```

Conclusion

This project successfully creates a realistic AI digital twin that can speak in my own voice through a sophisticated ComfyUI workflow. By combining cutting-edge models for image generation, voice cloning, and facial animation, the pipeline produces natural, cohesive talking-head videos that maintain visual consistency and accurate lip-synchronization.

The modular architecture allows for easy customization and future enhancements, while the detailed documentation ensures reproducibility and serves as a foundation for further development.